



BDF integrators for constrained mechanical systems on Lie groups

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ABSTRACT

Multistep methods of BDF type are the method-of-choice in several industrial multibody system simulation packages. In the present paper, BDF is applied to constrained systems in nonlinear configuration spaces with Lie group structure that allows, e.g., a representation of multibody systems with large rotations without singularities. The k -step Lie group integrator BLieDF avoids order reduction by a slightly perturbed argument of the exponential map for representing the nonlinearity of the numerical flow in the configuration space without any time-consuming re-parametrization.

This integrator is compared with multistep methods on Lie groups suggested by Faltinsen, Marthinsen and Munthe-Kaas (2001) and the advantages of the novel BLieDF integrator are shown.

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1. Introduction

In industrial simulation packages, the BDF method with step size and order control is the most common integrator for multibody systems [1]. The BDF method was first introduced as an implicit multistep method of order $p = k$ for ordinary differential equations by Curtiss and Hirschfelder [2]. BDF became famous for solving stiff differential equations by the work of Gear [3]. In the present paper these BDF integrators are applied to multibody systems with large rotations that are described without singularities by considering differential equations on manifolds with Lie group structure. The corresponding equations of motion form a differential–algebraic equation (DAE) of index 3.

Integrators for configuration spaces with Lie group structure have been considered by several authors. Crouch and Grossmann [4] investigated the numerical integration of ordinary differential equations on manifolds with explicit formulas for 3rd order multistep and Runge–Kutta methods by approximations directly in the Lie group G using several combinations of the exponential map. To reduce the number of exponential maps, Munthe-Kaas [5,6] uses approximations in the Lie algebra for Runge–Kutta methods on Lie groups. To obtain the high order in these Lie group formulations he solves the kinematic equations on the Lie group by an ODE that acts on the corresponding Lie algebra and therefore Lie brackets are involved. Celledoni, Marthinsen and Owren [7] tried to avoid Lie brackets and use a lower number of exponentials than Crouch and Grossmann [4] for the Runge–Kutta methods. It depends on the problem whether less exponential maps or less Lie brackets are computationally more efficient.

The aim of the present paper is to use only one exponential per time step and to reduce the number of Lie brackets for BDF Lie group integrators. Therefore ideas of Brüls and Cardona [8] are considered who have shown how to avoid time-consuming re-parametrizations of the manifold in generalized- α time integration. They reformulate the generalized- α

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method in terms of solution increments and combine linear updates of these increments with the exponential map to update the configuration variables such that the Lie group structure of the (nonlinear) configuration space is respected. In the present paper, an increment representation of BDF methods is used to get the BDF Lie group integrator BLieDF that has a small number of Lie brackets and evaluates only one exponential per time step. This integrator is applied to constrained mechanical systems that are described by Lie group differential–algebraic equations of index 3 for fixed step-sizes and convergence with order $p = k$ in all solution components is shown (for $2 \leq k \leq 4$).

In the unconstrained case, the application of multistep methods to differential equations on Lie groups has been considered by Faltinsen, Marthinsen and Munthe-Kaas [9], who use strategies of Munthe-Kaas [5,6]. In the present paper, this method is extended to Lie group DAEs of index 3 and denoted as Munthe-Kaas BDF Lie group integrator. In this method, local coordinates and re-parametrizations of the manifold are evaluated before every time step to handle the inherent nonlinearity of the Lie group formulations.

In terms of computing time the novel approach BLieDF to BDF time integration on Lie groups is clearly superior to existing multistep Lie group methods and has the potential to be implemented in industrial multibody system simulation software.

The paper is organized as follows. In Section 2, the BDF methods for ordinary differential equations are introduced, as well as some basics from Lie group theory. The BLieDF Lie group integrator for unconstrained and constrained mechanical systems is explained in Section 3. The next section considers the Munthe-Kaas BDF Lie group integrator. Section 5 compares both approaches for $2 \leq k \leq 4$. Afterwards the convergence with order $p = k$ is proved for $2 \leq k \leq 4$ for the BLieDF integrator for unconstrained (Section 6.1) and constrained (Section 6.2) mechanical systems. The final section is devoted to numerical tests for the Heavy top benchmark in the Lie group formulations $\mathbb{R}^3 \times SO(3)$ and $SE(3)$.

2. Basic concepts

2.1. Increment representation of BDF

BDF methods were originally proposed by Curtiss and Hirschfelder [2] for the time integration of first order ODEs. For the application to second order ODEs

$$\ddot{\mathbf{q}}(t) = \mathbf{a}(t, \mathbf{q}(t), \dot{\mathbf{q}}(t))$$

with $\mathbf{q} : [t_0, t_e] \rightarrow \mathbb{R}^N$, $\mathbf{a} : [t_0, t_e] \times \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}^N$, the equivalent first order system

$$\dot{\mathbf{q}}(t) = \mathbf{v}(t), \quad (1a)$$

$$\dot{\mathbf{v}}(t) = \mathbf{a}(t, \mathbf{q}(t), \mathbf{v}(t)) \quad (1b)$$

with new variables $\mathbf{v} : [t_0, t_e] \rightarrow \mathbb{R}^N$ is considered. For fixed step size h , the time step $t_n \rightarrow t_{n+1} = t_n + h$ of a k -step BDF discretization for the partitioned system (1) is given by

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{q}_{n+1-i} = \mathbf{v}_{n+1}, \quad (2a)$$

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{v}_{n+1-i} = \mathbf{a}(t_{n+1}, \mathbf{q}_{n+1}, \mathbf{v}_{n+1}) \quad (2b)$$

with known variables $\mathbf{q}_{n+1-i}$, $\mathbf{v}_{n+1-i}$, ($i = 1, \dots, k$). Eq. (2) defines solutions $\mathbf{q}_{n+1}$ and $\mathbf{v}_{n+1}$. The method is initialized by starting values $\mathbf{q}_i \approx \mathbf{q}(t_i)$ and $\mathbf{v}_i \approx \mathbf{v}(t_i)$, ($i = 0, \dots, k-1$). The algorithmic parameters α_i , ($i = 0, 1, \dots, k$) satisfy the order conditions

$$\sum_{i=0}^k \alpha_i = 0 \text{ and } \sum_{i=0}^k \alpha_i \frac{(k-i)^l}{k^{l-1}} = l, \quad (l = 1, \dots, k). \quad (3)$$

For $2 \leq k \leq 4$ these parameters are given by

$$\begin{aligned} k=2 : \alpha_0 &= \frac{3}{2}, \quad \alpha_1 = -2, \quad \alpha_2 = \frac{1}{2}, \\ k=3 : \alpha_0 &= \frac{11}{6}, \quad \alpha_1 = -3, \quad \alpha_2 = \frac{3}{2}, \quad \alpha_3 = -\frac{1}{3}, \\ k=4 : \alpha_0 &= \frac{25}{12}, \quad \alpha_1 = -4, \quad \alpha_2 = 3, \quad \alpha_3 = -\frac{4}{3}, \quad \alpha_4 = \frac{1}{4}. \end{aligned}$$

In [10] it is shown, that method (2) for $1 \leq k \leq 6$ has an order of convergence $p = k$ and is zero-stable. The method is A-stable for $k \leq 2$ and A(α)-stable for $3 \leq k \leq 6$.

In the context of multibody system dynamics, the variables $\mathbf{q}$, $\mathbf{v}$ and $\mathbf{a}$ represent position, velocity and acceleration variables, respectively. The partitioned system (1) summarizes the kinematic equations (1a) and the dynamical equations (1b). These dynamical equations result from the equilibrium conditions

$$\mathbf{M}(\mathbf{q}(t))\dot{\mathbf{v}}(t) + \mathbf{g}(t, \mathbf{q}(t), \mathbf{v}(t)) = \mathbf{0} \quad (4)$$

for forces and moments by left multiplication with the inverse of the symmetric, positive definite mass matrix $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{N \times N}$:

$$\mathbf{a}(t, \mathbf{q}(t), \mathbf{v}(t)) := (\mathbf{M}(\mathbf{q}(t)))^{-1}(-\mathbf{g}(t, \mathbf{q}(t), \mathbf{v}(t))). \quad (5)$$

The vector valued function $\mathbf{g} : [t_0, t_e] \times \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}^N$ in the equilibrium conditions (4) denotes a force vector. As before, the function $\mathbf{a}(t, \mathbf{q}, \mathbf{v})$ represents the accelerations.

The main interest of the present paper is in model equations in *nonlinear* configuration spaces with Lie group structure. As seen in Section 2.2, the equilibrium conditions (4) and the dynamical equations (1b) keep their structure also in this more general setting but the kinematic equations (1a) get a substantially more complex form. For this framework, it is useful to rewrite the BDF discretization (2a) for the position coordinates $\mathbf{q}$ in the increment form

$$\frac{1}{h} \sum_{i=1}^k \gamma_i \mathbf{d}\mathbf{q}_{n+1-i} = \mathbf{v}_{n+1} \quad (6)$$

with the one-step update $\mathbf{q}_{n+1} = \mathbf{q}_n + \mathbf{d}\mathbf{q}_n$ and vectors

$$\mathbf{d}\mathbf{q}_{n+1-i} := \mathbf{q}_{n+2-i} - \mathbf{q}_{n+1-i}, \quad (i = 2, \dots, k),$$

that are considered to be given in time step $t_n \rightarrow t_{n+1}$. Since

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{q}_{n+1-i} = \frac{1}{h} \sum_{i=1}^k \gamma_i \mathbf{d}\mathbf{q}_{n+1-i},$$

the new parameters are defined by

$$\gamma_i := \sum_{j=0}^{i-1} \alpha_j, \quad (i = 1, \dots, k),$$

where

$$\begin{aligned} k=2 : \gamma_1 &= \frac{3}{2}, \quad \gamma_2 = -\frac{1}{2}, \\ k=3 : \gamma_1 &= \frac{11}{6}, \quad \gamma_2 = -\frac{7}{6}, \quad \gamma_3 = \frac{1}{3}, \\ k=4 : \gamma_1 &= \frac{25}{12}, \quad \gamma_2 = -\frac{23}{12}, \quad \gamma_3 = \frac{13}{12}, \quad \gamma_4 = -\frac{1}{4}. \end{aligned}$$

All in all, a k -step BDF integrator for (1) may be written in the form

$$\mathbf{q}_{n+1} = \mathbf{q}_n + \mathbf{d}\mathbf{q}_n, \quad (7a)$$

$$\frac{1}{h} \sum_{i=1}^k \gamma_i \mathbf{d}\mathbf{q}_{n+1-i} = \mathbf{v}_{n+1}, \quad (7b)$$

$$\mathbf{M}(\mathbf{q}_{n+1}) \frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{v}_{n+1-i} = -\mathbf{g}(t_{n+1}, \mathbf{q}_{n+1}, \mathbf{v}_{n+1}). \quad (7c)$$

For practical implementations, it is not efficient to calculate in (5) the inverse of matrix $\mathbf{M}(\mathbf{q}_{n+1})$. Therefore Eq. (2b) is multiplied with the mass matrix $\mathbf{M}(\mathbf{q}_{n+1})$ resulting in the equivalent equation (7c).

2.2. Multibody systems: Configuration spaces with Lie group structure

To describe a rigid body with large rotations in 3D spaces, a representation without singularities is not possible by coordinates in a linear space. Therefore the position coordinates $\mathbf{q}$ in (1) must be coordinates of a nonlinear configuration space with e.g. a Lie group structure.

A Lie group G is a differentiable manifold with neutral element e , a differentiable group operation $\circ : G \times G \rightarrow G$ and a differentiable inverse mapping $q \mapsto q^{-1}$ ($q \circ q^{-1} = e$). Following [11], the focus is on matrix Lie groups, which are defined as subgroups of the general linear group $GL(n_G)$. See [12] for a more general introduction to Lie group theory.

The configuration of a multibody system at time t can be characterized by the position coordinates $q(t) : [t_0, t_e] \rightarrow G$ that describe position and orientation of all bodies in the system. As already discussed in [8,13], a continuously differentiable function $q(t)$ with $q(t_0) \in G$ will remain in a Lie group G if and only if its time derivative $\dot{q}(t)$ is in the tangent space $T_q G$ at the point $q = q(t) : \dot{q}(t) \in T_{q(t)} G$, ($t \geq t_0$). The tangent space at the neutral element e defines the Lie algebra $\mathfrak{g} := T_e G$. As a linear space, it is isomorphic to a finite dimensional linear space $\mathbb{R}^N$ with an invertible linear mapping $(\bullet) : \mathbb{R}^N \rightarrow \mathfrak{g}$, $\mathbf{v} \mapsto \tilde{\mathbf{v}}$.

The group structure of G makes it possible to represent the elements of $T_q G$ at any $q \in G$ by the elements $\tilde{\mathbf{v}}$ of the Lie algebra: The left translation $L_q : G \rightarrow G$, $y \mapsto L_q(y) := q \circ y$ defines a bijection in G . Its derivative $DL_q(y)$ at $y = e$ represents the corresponding bijection between the tangent spaces $\mathfrak{g} := T_e G$ and $T_q G$, i.e.,

$$T_q G = \{DL_q(e) \cdot \tilde{\mathbf{v}} : \tilde{\mathbf{v}} \in \mathfrak{g}\} = \{DL_q(e) \cdot \tilde{\mathbf{v}} : \mathbf{v} \in \mathbb{R}^N\}.$$

In this framework, the kinematic equations are given by [8]

$$\dot{q}(t) = DL_q(e) \cdot \tilde{\mathbf{v}}(t) \quad (8)$$

with velocity coordinates $\mathbf{v} : [t_0, t_e] \rightarrow \mathbb{R}^N$.

The exponential map $\exp : \mathfrak{g} \rightarrow G$ maps the Lie algebra $\mathfrak{g}$ to the Lie group G . In a neighbourhood of $t = t_m$, the solution of the kinematic equations (8) is given by

$$q(t) = q(t_m) \circ \exp(\tilde{\Omega}_m(t - t_m)), \quad t \in \mathbb{R} \quad (9)$$

with $\tilde{\Omega}_m(t) \in \mathfrak{g}$ solving the initial value problem

$$\dot{\tilde{\Omega}}_m(t) = \text{dexp}_{-\tilde{\Omega}_m(t)}^{-1} \tilde{\mathbf{v}}(t), \quad \tilde{\Omega}_m(0) = \mathbf{0} \quad (10)$$

with $\text{dexp}_{\tilde{\Omega}}^{-1}$ denoting the inverse of the trivialized derivative $\text{dexp}_{\tilde{\Omega}}$ of the exponential map, see [11, Theorem IV.7.1]. In a neighbourhood of $\tilde{\Omega} = \mathbf{0}$, this inverse is given by

$$\text{dexp}_{\tilde{\Omega}}^{-1} \tilde{\mathbf{v}} = \sum_{k=0}^{B_k} \frac{B_k}{k!} \text{ad}_{\tilde{\Omega}}^k(\tilde{\mathbf{v}}), \quad (\tilde{\mathbf{v}} \in \mathfrak{g}), \quad (11)$$

see [11,14]. The expansion (11) involves the Bernoulli numbers B_k and the adjoint operator $\text{ad}_{\tilde{\Omega}} \tilde{\mathbf{v}} = [\tilde{\Omega}, \tilde{\mathbf{v}}]$ with $\text{ad}_{\tilde{\Omega}}^0 \tilde{\mathbf{v}} = \tilde{\mathbf{v}}$ and $\text{ad}_{\tilde{\Omega}}^k \tilde{\mathbf{v}} = \text{ad}_{\tilde{\Omega}}(\text{ad}_{\tilde{\Omega}}^{k-1} \tilde{\mathbf{v}})$ for $k \geq 1$ and the Lie bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ which is a bilinear skew symmetric operation on $\mathfrak{g}$, that would vanish when the Lie group G is replaced by a linear configuration space. The solution of (10) can be calculated by the Magnus expansion [14] and approximated for $t = t_m + h$ by (see [15])

$$\tilde{\Omega}_m(h) = h\tilde{\mathbf{v}}(t_m) + \frac{h^2}{2}\dot{\tilde{\mathbf{v}}}(t_m) + \frac{h^3}{6}\ddot{\tilde{\mathbf{v}}}(t_m) + \frac{h^4}{24}\ddot{\tilde{\mathbf{v}}}(t_m) + \frac{h^3}{12}[\tilde{\mathbf{v}}(t_m), \dot{\tilde{\mathbf{v}}}(t_m)] + \frac{h^4}{24}[\tilde{\mathbf{v}}(t_m), \ddot{\tilde{\mathbf{v}}}(t_m)] + \mathcal{O}(h^5), \quad (12)$$

$h \rightarrow 0$. For a more detailed discussion on Lie groups in multibody dynamics see [13,15,16].

3. BLieDF

3.1. BLieDF integrator for unconstrained mechanical systems

For unconstrained mechanical systems on Lie groups, the equation $\dot{\mathbf{q}} = \mathbf{v}$ in (1) needs to be replaced by the kinematic equations (8). With acceleration vector (5), the equations of motion become

$$\dot{q}(t) = DL_{q(t)}(e) \cdot \tilde{\mathbf{v}}(t), \quad (13a)$$

$$\mathbf{M}(q(t))\dot{\mathbf{v}}(t) = -\mathbf{g}(t, q(t), \mathbf{v}(t)) \quad (13b)$$

with position coordinates $q(t) \in G$, velocity coordinates $\mathbf{v}(t) \in \mathbb{R}^N$, positive definite, symmetric mass matrix $\mathbf{M}(q(t))$ and force vector $\mathbf{g}(t, q(t), \mathbf{v}(t))$.

Following the strategy of Bruls and Cardona [8], the traditional update of configuration variables in the kinematic equations is substituted by updates of solution increments. Eq. (9) with $m = n$ and $t = t_{n+1} = t_n + h$, shows that $\tilde{\Omega}_n(h) \in \mathfrak{g}$ is the analogue to the solution increment $\mathbf{dq}_n \in \mathbb{R}^N$ in (7a):

$$q(t_{n+1}) = q(t_n) \circ \exp(\tilde{\Omega}_n(h)).$$

Therefore the update of the configuration variables in time step $t_n \rightarrow t_{n+1}$ is defined by

$$q_{n+1} = q_n \circ \exp(\tilde{\omega}_0^{(n)})$$

with approximations $\tilde{\omega}_i^{(n)} \approx \tilde{\Omega}_n((1-i)h)$ for $i = 0, \dots, k$. For local truncation error analysis the weighted sum $\sum_{i=1}^k \gamma_i \tilde{\Omega}_{n+1-i}(h)$ is considered, see (7b):

Lemma 1. If $\mathbf{v}(t)$ is k -times continuously differentiable and $k \leq 4$ then

$$\frac{1}{h} \sum_{i=1}^k \gamma_i \tilde{\Omega}_{n+1-i}(h) = \tilde{\mathbf{v}}(t_{n+1}) + \frac{h^2}{12} [\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_n)] + \mathcal{O}(h^k), \quad (h \rightarrow 0).$$

Proof. Taylor expansion for Eq. (12) with $m = n + 1 - i$ leads to

$$\begin{aligned} \tilde{\Omega}_{n+1-i}(h) &= h\tilde{\mathbf{v}}(t_n) + \left(\frac{3}{2} - i\right) h^2 \dot{\tilde{\mathbf{v}}}(t_n) + \left(\frac{7}{6} - \frac{3}{2}i + \frac{1}{2}i^2\right) h^3 \ddot{\tilde{\mathbf{v}}}(t_n) + \frac{h^3}{12} [\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_n)] \\ &\quad + \left(\frac{5}{8} - \frac{7}{6}i + \frac{3}{4}i^2 - \frac{1}{6}i^3\right) h^4 \ddot{\tilde{\mathbf{v}}}(t_n) + \left(\frac{1}{8} - \frac{1}{12}i\right) h^4 [\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_n)] + \mathcal{O}(h^5). \end{aligned} \quad (14)$$

The first two terms of the weighted sum $\frac{1}{h} \sum_{i=1}^k \gamma_i \tilde{\Omega}_{n+1-i}(h)$ are therefore given by

$$\begin{aligned} \frac{1}{h} \sum_{i=1}^k \gamma_i h \tilde{\mathbf{v}}(t_n) &= \frac{1}{h} \sum_{i=1}^k \left(\sum_{j=0}^{i-1} \alpha_j \right) h \tilde{\mathbf{v}}(t_n) = \sum_{j=0}^{k-1} \alpha_j \sum_{i=j+1}^k \tilde{\mathbf{v}}(t_n) = \sum_{j=0}^{k-1} \alpha_j (k-j) \tilde{\mathbf{v}}(t_n) \\ &= \left(\sum_{j=0}^k \alpha_j (k-j) \right) \tilde{\mathbf{v}}(t_n) = \tilde{\mathbf{v}}(t_n), \end{aligned}$$

see (3) with $l = 1$, and by

$$\begin{aligned} &\frac{1}{h} \sum_{i=1}^k \gamma_i \left(\frac{3}{2} - i \right) h^2 \dot{\tilde{\mathbf{v}}}(t_n) \\ &= h \sum_{j=0}^{k-1} \alpha_j \left(\sum_{i=j+1}^k \left(\frac{3}{2} - i \right) \dot{\tilde{\mathbf{v}}}(t_n) \right) = h \left(\sum_{j=0}^{k-1} \frac{3}{2} \alpha_j (k-j) - \sum_{j=0}^{k-1} \alpha_j \left(\frac{k(k+1)}{2} - \frac{j(j+1)}{2} \right) \right) \dot{\tilde{\mathbf{v}}}(t_n) \\ &= h \left(\sum_{j=0}^k \frac{3}{2} \alpha_j (k-j) + \frac{1}{2} \sum_{j=0}^k \alpha_j (k-j)^2 - \left(k + \frac{1}{2} \right) \sum_{j=0}^k \alpha_j (k-j) \right) \dot{\tilde{\mathbf{v}}}(t_n) \\ &= h \left(\frac{3}{2} + \frac{1}{2} 2k - \left(k + \frac{1}{2} \right) \right) \dot{\tilde{\mathbf{v}}}(t_n) = h \dot{\tilde{\mathbf{v}}}(t_n) \end{aligned}$$

if $k \geq 2$, see (3). Proceeding in this way, the order conditions (3) finally yield

$$\frac{1}{h} \sum_{i=1}^k \gamma_i \tilde{\Omega}_{n+1-i}(h) = \tilde{\mathbf{v}}(t_n) + h \dot{\tilde{\mathbf{v}}}(t_n) + \frac{h^2}{2} \ddot{\tilde{\mathbf{v}}}(t_n) + \frac{h^3}{6} \ddot{\tilde{\mathbf{v}}}(t_n) + \frac{h^2}{12} [\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_n)] + \frac{h^3}{12} [\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_n)] + \mathcal{O}(h^k)$$

for $k \leq 4$. The proposition is verified by Taylor expansion of $\tilde{\mathbf{v}}(t_{n+1}) = \tilde{\mathbf{v}}(t_n + h)$, taking into account

$$\frac{h^2}{12} [\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_n)] + \frac{h^3}{12} [\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_n)] = \frac{h^2}{12} [\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_{n+1})] + \mathcal{O}(h^4). \quad \blacksquare$$

The Lie bracket $[\tilde{\mathbf{v}}(t_n), \dot{\tilde{\mathbf{v}}}(t_{n+1})]$ in Lemma 1 would vanish if the Lie group G would be replaced by a linear configuration space. In the k -step Lie group integrator, it has to be considered by a correction term $\mathbf{L}_k$ that needs to fulfil

$$\mathbf{L}_1(\mathbf{v}(t); h) = \mathcal{O}(h), \quad (15a)$$

$$\mathbf{L}_2(\mathbf{v}(t), \mathbf{v}(t-h); h) = \mathcal{O}(h^2), \quad (15b)$$

$$\mathbf{L}_3(\mathbf{v}(t), \mathbf{v}(t-h), \mathbf{v}(t-2h); h) = \frac{h^2}{12} [\tilde{\mathbf{v}}(t), \dot{\tilde{\mathbf{v}}}(t)] + \mathcal{O}(h^3), \quad (15c)$$

$$\mathbf{L}_4(\mathbf{v}(t), \mathbf{v}(t-h), \mathbf{v}(t-2h), \mathbf{v}(t-3h); h) = \frac{h^2}{12} [\tilde{\mathbf{v}}(t), \dot{\tilde{\mathbf{v}}}(t+h)] + \mathcal{O}(h^4), \quad (15d)$$

see Lemma 1. The velocity coordinates $\mathbf{v}$ are elements of the linear space $\mathbb{R}^N$, therefore the classical BDF update (7c) is applied. The k -step BLieDF method for (13) in time step $t_n \rightarrow t_{n+1} := t_n + h$ may be written as

$$q_{n+1} = q_n \circ \exp(\tilde{\omega}_0^{(n)}), \quad (16a)$$

$$\frac{1}{h} \sum_{i=1}^k \gamma_i \omega_0^{(n+1-i)} = \mathbf{v}_{n+1} + \mathbf{L}_k(\mathbf{v}_n, \mathbf{v}_{n-1}, \dots, \mathbf{v}_{n-k+1}; h), \quad (16b)$$

$$\mathbf{M}(q_{n+1}) \frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{v}_{n+1-i} = -\mathbf{g}(t_{n+1}, q_{n+1}, \mathbf{v}_{n+1}) \quad (16c)$$

with a suitable correction term $\mathbf{L}_k$ that satisfies (15). Using difference approximations

$$\begin{aligned} \dot{\mathbf{v}}(t) &= \frac{6\mathbf{v}(t) - 11\mathbf{v}(t-h) + 5\mathbf{v}(t-2h)}{h} + \mathcal{O}(h), \\ \dot{\mathbf{v}}(t+h) &= \frac{\frac{19}{3}\mathbf{v}(t) - \frac{31}{2}\mathbf{v}(t-h) + 13\mathbf{v}(t-2h) - \frac{23}{6}\mathbf{v}(t-3h)}{h} + \mathcal{O}(h^2) \end{aligned}$$

with $\mathbf{v}(t-ih)$ being substituted by the numerical solutions $\mathbf{v}_{n-i}$ for $i = 0, 1, 2, 3$, the terms

$$\mathbf{L}_1 = \mathbf{L}_2 := \mathbf{0}, \quad (17a)$$

$$\tilde{\mathbf{L}}_3 := \frac{h}{12} [\tilde{\mathbf{v}}_n, 6\tilde{\mathbf{v}}_n - 11\tilde{\mathbf{v}}_{n-1} + 5\tilde{\mathbf{v}}_{n-2}], \quad (17b)$$

$$\tilde{\mathbf{L}}_4 := \frac{h}{12} [\tilde{\mathbf{v}}_n, \frac{19}{3}\tilde{\mathbf{v}}_n - \frac{31}{2}\tilde{\mathbf{v}}_{n-1} + 13\tilde{\mathbf{v}}_{n-2} - \frac{23}{6}\tilde{\mathbf{v}}_{n-3}] \quad (17c)$$

proved to be useful in practical computations. In future research an optimal choice of the terms $\mathbf{L}_k$ will be considered.

The integrator is initialized by $q_{k-1}, \omega_0^{(j)}, \mathbf{v}_j$ with

$$q_{k-1} \approx q(t_{k-1}), \quad q(t_{j+1}) \approx q(t_j) \circ \exp(\tilde{\omega}_0^{(j)}), \quad (j = 0, \dots, k-2), \quad \mathbf{v}_j \approx \mathbf{v}(t_j), \quad (j = 0, \dots, k-1).$$

Note, that this initialization requires $k-1$ evaluations of the inverse of the exponential map to get approximations $\omega_0^{(j)}$, ($j = 0, \dots, k-2$), for the first time step.

3.2. BLieDF integrator for constrained mechanical systems

The BLieDF integrator (16) is applied to constrained mechanical systems. The equations of motion are given as differential-algebraic equations in the index-3 formulation [8,16]

$$\dot{q}(t) = DL_{q(t)}(e) \cdot \tilde{\mathbf{v}}(t), \quad (18a)$$

$$\mathbf{M}(q(t))\dot{\mathbf{v}}(t) = -\mathbf{g}(t, q(t), \mathbf{v}(t)) - \mathbf{B}^T(q(t))\boldsymbol{\lambda}(t), \quad (18b)$$

$$\mathbf{0} = \Phi(q(t)) \quad (18c)$$

with position coordinates $q(t) \in G$, velocity coordinates $\mathbf{v}(t) \in \mathbb{R}^N$, positive definite, symmetric mass matrix $\mathbf{M}(q(t))$, force vector $\mathbf{g}(t, q(t), \mathbf{v}(t))$, Lagrange multipliers $\boldsymbol{\lambda}(t) \in \mathbb{R}^l$, $l \leq N$ linearly independent holonomic constraints (18c) and constraint gradients $\mathbf{B}(q(t))$ with

$$D\Phi(q) \cdot (DL_q(e) \cdot \tilde{\mathbf{w}}) = \mathbf{B}(q)\mathbf{w}, \quad (\mathbf{w} \in \mathbb{R}^N). \quad (19)$$

To apply (16) to these equations, the holonomic constraint (18c) and the Lagrange multipliers $\boldsymbol{\lambda}$ need to be added. The k -step BLieDF integrator (16) for constrained mechanical systems in the index-3 formulation (18) is for the time step $t_n \rightarrow t_{n+1} := t_n + h$ defined by

$$q_{n+1} = q_n \circ \exp(\tilde{\omega}_0^{(n)}), \quad (20a)$$

$$\frac{1}{h} \sum_{i=1}^k \gamma_i \omega_0^{(n+1-i)} = \mathbf{v}_{n+1} + \mathbf{L}_k(\mathbf{v}_n, \mathbf{v}_{n-1}, \dots, \mathbf{v}_{n-k+1}; h), \quad (20b)$$

$$\mathbf{M}(q_{n+1}) \frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{v}_{n+1-i} = -\mathbf{g}(t_{n+1}, q_{n+1}, \mathbf{v}_{n+1}) - \mathbf{B}^T(q_{n+1})\boldsymbol{\lambda}_{n+1}, \quad (20c)$$

$$\mathbf{0} = \Phi(q_{n+1}). \quad (20d)$$

4. Munthe-Kaas BDF Lie group integrator

Faltinsen, Marthinsen and Munthe-Kaas [9] consider linear multistep methods for first order ODEs on Lie groups. In linear spaces, these methods coincide with their well known counterparts from classical ODE theory. In the application of BDF to the partitioned system (13), they discretize the dynamical equations (13b) by

$$\mathbf{M}(q_{n+1}) \frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{v}_{n+1-i} = -\mathbf{g}(t_{n+1}, q_{n+1}, \mathbf{v}_{n+1}),$$

see (7c). For the kinematic equation (13a), Faltinsen, Marthinsen and Munthe-Kaas [9] propose to approximate in (10) the inverse $\text{dexp}_{-\tilde{\omega}}^{-1}(\tilde{\mathbf{v}})$ of the trivialized derivative of the exponential map by

$$\begin{aligned}(\text{dexp}_2^{-1})_{-\tilde{\omega}}(\tilde{\mathbf{v}}) &:= \tilde{\mathbf{v}}, \\(\text{dexp}_3^{-1})_{-\tilde{\omega}}(\tilde{\mathbf{v}}) &:= (\text{dexp}_2^{-1})_{-\tilde{\omega}}(\tilde{\mathbf{v}}) + \frac{1}{2}[\tilde{\omega}, \tilde{\mathbf{v}}], \\(\text{dexp}_4^{-1})_{-\tilde{\omega}}(\tilde{\mathbf{v}}) &:= (\text{dexp}_3^{-1})_{-\tilde{\omega}}(\tilde{\mathbf{v}}) + \frac{1}{12}[\tilde{\omega}, [\tilde{\omega}, \tilde{\mathbf{v}}]],\end{aligned}$$

see (11). The BDF discretization of this approximation is given by

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \tilde{\omega}_i^{(n)} = (\text{dexp}_k^{-1})_{-\tilde{\omega}_0^{(n)}}(\tilde{\mathbf{v}}_{n+1})$$

with $\tilde{\omega}_i^{(n)}$ denoting the local coordinates of q_{n+1-i} w.r.t. q_n in the sense that

$$q_{n+1-i} = q_n \circ \exp(\tilde{\omega}_i^{(n)}), \quad (i = 0, 1, \dots, k). \quad (21)$$

The comparison of (21) with (9) shows $\tilde{\omega}_i^{(n)} \approx \tilde{\Omega}_n((1-i)h)$, since $q_n \approx q(t_n)$ and $q_{n+1-i} \approx q(t_{n+1-i})$ for $i = 0, 1, \dots, k$. Furthermore Eq. (9) for $m = n$ and Eq. (9) for $m = n-1$ result in a neighbourhood of $t = t_n$ in

$$q(t_n) \circ \exp(\tilde{\Omega}_n(t - t_n)) = q(t) = q(t_{n-1}) \circ \exp(\tilde{\Omega}_{n-1}(t - t_{n-1}))$$

and therefore

$$\exp(\tilde{\Omega}_n((1-i)h)) = q(t_n)^{-1} \circ q(t_{n-1}) \circ \exp(\tilde{\Omega}_{n-1}((2-i)h)) = \exp(-\tilde{\Omega}_{n-1}(h)) \circ \exp(\tilde{\Omega}_{n-1}((2-i)h)).$$

In time step $t_n \rightarrow t_{n+1}$, the approximations $-\tilde{\omega}_0^{(n-1)} \approx -\tilde{\Omega}_{n-1}(h)$ and $\tilde{\omega}_{i-1}^{(n-1)} \approx \tilde{\Omega}_{n-1}((2-i)h)$ for $i = 1, \dots, k$, are known and allow to evaluate $\tilde{\omega}_i^{(n)} \approx \tilde{\Omega}_n((1-i)h)$ with

$$\exp(\tilde{\omega}_i^{(n)}) \approx \exp(-\tilde{\omega}_0^{(n-1)}) \circ \exp(\tilde{\omega}_{i-1}^{(n-1)}), \quad (i = 1, \dots, k).$$

Formally, the product of exponentials is determined by the Baker–Campbell–Hausdorff formula. In the practical implementation, this formula is substituted by approximations BCH_k , see [9,12]:

$$\begin{aligned}\text{BCH}_2(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2) &:= \tilde{\mathbf{v}}_1 + \tilde{\mathbf{v}}_2, \\ \text{BCH}_3(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2) &:= \text{BCH}_2(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2) + \frac{1}{2}[\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2], \\ \text{BCH}_4(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2) &:= \text{BCH}_3(\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2) + \frac{1}{12}[\tilde{\mathbf{v}}_1 - \tilde{\mathbf{v}}_2, [\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2]].\end{aligned}$$

For the resulting k -step Munthe-Kaas BDF Lie group integrator for unconstrained systems (13), time step $t_n \rightarrow t_{n+1} := t_n + h$ is given by

$$q_{n+1} = q_n \circ \exp(\tilde{\omega}_0^{(n)}), \quad (22a)$$

$$\tilde{\omega}_i^{(n)} = \text{BCH}_k(-\tilde{\omega}_0^{(n-1)}, \tilde{\omega}_{i-1}^{(n-1)}), \quad i = 1, \dots, k, \quad (22b)$$

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \tilde{\omega}_i^{(n)} = (\text{dexp}_k^{-1})_{-\tilde{\omega}_0^{(n)}}(\tilde{\mathbf{v}}_{n+1}), \quad (22c)$$

$$\mathbf{M}(q_{n+1}) \frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{v}_{n+1-i} = -\mathbf{g}(t_{n+1}, q_{n+1}, \mathbf{v}_{n+1}) \quad (22d)$$

with starting values

$$q_{k-1} \approx q(t_{k-1}), \quad q(t_{k-j}) \approx q(t_{k-1}) \circ \exp(\tilde{\omega}_j^{(k-1)}), \quad \mathbf{v}_j \approx \mathbf{v}(t_j), \quad (j = 0, \dots, k-1).$$

For the application to constrained systems with equations of motion (18) the equilibrium conditions (22d) are substituted by

$$\mathbf{M}(q_{n+1}) \frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{v}_{n+1-i} = -\mathbf{g}(t_{n+1}, q_{n+1}, \mathbf{v}_{n+1}) - \mathbf{B}^T(q_{n+1}) \boldsymbol{\lambda}_{n+1}, \quad (23a)$$

$$\mathbf{0} = \Phi(q_{n+1}). \quad (23b)$$

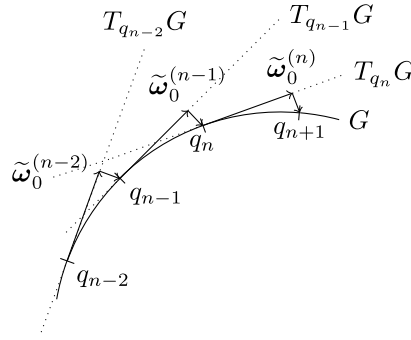


Fig. 1. BLieDF method: Different tangent spaces.

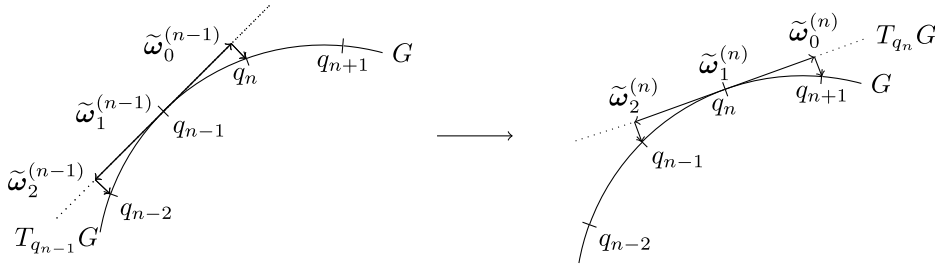


Fig. 2. Munthe-Kaas BDF Lie group method: Re-parametrization into another tangent space.

5. Comparison of the two BDF Lie group integrators for $2 \leq k \leq 4$

The BDF Lie group integrators (16) and (22) address the nonlinear structure of the configuration space in different ways. In method (16) the increment BDF form (6) is directly applied, so that elements $DL_{q_{n+1-i}}(e) \cdot \tilde{\omega}_0^{(n+1-i)}$ for $i = 1, \dots, k$ from different tangent spaces $T_{q_{n+1-i}}G$ are used to update the configuration variables, see Fig. 1. The nonlinear structure of the configuration space is considered by the correction term L_k in (16b).

In contrast, method (22) is based on the classical BDF form (2) to solve for $m = n$ the auxiliary ODE (10) that is defined in a linear space. The numerical solutions $\tilde{\omega}_i^{(n-1)} \in \mathfrak{g}$ with $DL_{q_{n-1}}(e) \cdot \tilde{\omega}_i^{(n-1)} \in T_{q_{n-1}}G$ for $i = 0, \dots, k-1$ of the previous time step are transformed to $\tilde{\omega}_i^{(n)} \in \mathfrak{g}$ with $DL_{q_n}(e) \cdot \tilde{\omega}_i^{(n)} \in T_{q_n}G$ for $i = 1, \dots, k$, see Fig. 2. This re-parametrization by BCH_k -formulas and the approximations dexp_k^{-1} reflect the nonlinearity of the configuration space and its Lie group structure.

The classical BDF discretizations in the velocity coordinates $\mathbf{v}$ (Eqs. (16c) and (22d)) are equivalent for both methods (16) and (22), as well as their extension to constrained systems.

For $k = 2$, both methods (16) and (22) can be written as

$$\begin{aligned} q_{n+1} &= q_n \circ \exp(\tilde{\omega}_0^{(n)}), \\ \frac{1}{h} \left(\frac{3}{2} \tilde{\omega}_0^{(n)} - \frac{1}{2} \tilde{\omega}_0^{(n-1)} \right) &= \mathbf{v}_{n+1}, \\ \mathbf{M}(q_{n+1}) \frac{1}{h} \left(\frac{3}{2} \mathbf{v}_{n+1} - 2\mathbf{v}_n + \frac{1}{2} \mathbf{v}_{n-1} \right) &= -\mathbf{g}(t_{n+1}, q_{n+1}, \mathbf{v}_{n+1}). \end{aligned}$$

This shows, that the integrators BLieDF (16) and Munthe-Kaas BDF (22) are equivalent for $k = 2$.

For $k > 2$, the updates (16a), (22a) of the configuration variables and the equilibrium conditions (16c), (22d) are still identical for both types of BDF Lie group integrators, but the BDF discretization in (16b) is different from the one in (22b)–(22c). For BLieDF with $k = 3$, the equation

$$\frac{1}{h} \left(\frac{11}{6} \tilde{\omega}_0^{(n)} - \frac{7}{6} \tilde{\omega}_0^{(n-1)} + \frac{1}{3} \tilde{\omega}_0^{(n-2)} \right) = \tilde{\mathbf{v}}_{n+1} + \frac{h}{12} [\tilde{\mathbf{v}}_n, 6\tilde{\mathbf{v}}_n - 11\tilde{\mathbf{v}}_{n-1} + 5\tilde{\mathbf{v}}_{n-2}] \quad (24)$$

is obtained. The corresponding Munthe-Kaas BDF Lie group integrator makes use of

$$\begin{aligned} \tilde{\omega}_1^{(n)} &= \text{BCH}_3(-\tilde{\omega}_0^{(n-1)}, \tilde{\omega}_0^{(n-1)}) = \tilde{\mathbf{0}}, \\ \tilde{\omega}_2^{(n)} &= \text{BCH}_3(-\tilde{\omega}_0^{(n-1)}, \tilde{\omega}_1^{(n-1)}) = \text{BCH}_3(-\tilde{\omega}_0^{(n-1)}, \tilde{\mathbf{0}}) = -\tilde{\omega}_0^{(n-1)} \end{aligned}$$

and

$$\begin{aligned}\tilde{\omega}_3^{(n)} &= -\tilde{\omega}_0^{(n-1)} + \tilde{\omega}_2^{(n-1)} - \frac{1}{2} [\tilde{\omega}_0^{(n-1)}, \tilde{\omega}_2^{(n-1)}], \\ \frac{1}{h} \left(\frac{11}{6} \tilde{\omega}_0^{(n)} + \frac{3}{2} \tilde{\omega}_2^{(n)} - \frac{1}{3} \tilde{\omega}_3^{(n)} \right) &= \tilde{\mathbf{v}}_{n+1} + \frac{1}{2} [\tilde{\omega}_0^{(n)}, \tilde{\mathbf{v}}_{n+1}],\end{aligned}$$

i.e.

$$\frac{1}{h} \left(\frac{11}{6} \tilde{\omega}_0^{(n)} - \frac{7}{6} \tilde{\omega}_0^{(n-1)} - \frac{1}{3} \tilde{\omega}_2^{(n-1)} \right) = \tilde{\mathbf{v}}_{n+1} + \frac{1}{2} [\tilde{\omega}_0^{(n)}, \tilde{\mathbf{v}}_{n+1}] - \frac{1}{6h} [\tilde{\omega}_0^{(n-1)}, \tilde{\omega}_2^{(n-1)}]. \quad (25)$$

The BLieDF integrator (24) only requires the evaluation of one Lie bracket per time step with explicitly given arguments. On the other hand the Munthe-Kaas BDF Lie group integrator (25) uses two Lie brackets. One of these Lie brackets must be evaluated in every Newton step, because $\tilde{\omega}_0^{(n)}$ and $\mathbf{v}_{n+1}$ are involved. This is why the first method (24) should be more efficient in terms of numerical effort per time step. Techniques for reducing the numerical effort in the evaluation or approximation of Lie brackets are discussed, e.g., in [17].

For $k = 4$, the BLieDF discretization (16b) becomes

$$\frac{1}{h} \sum_{i=1}^4 \gamma_i \tilde{\omega}_0^{(n+1-i)} = \tilde{\mathbf{v}}_{n+1} + \frac{h}{12} \left[\tilde{\mathbf{v}}_n, \frac{19}{3} \mathbf{v}_n - \frac{31}{2} \mathbf{v}_{n-1} + 13 \mathbf{v}_{n-2} - \frac{23}{6} \mathbf{v}_{n-3} \right]. \quad (26)$$

The Munthe-Kaas BDF Lie group discretization (22b)–(22c) is given by

$$\begin{aligned}\tilde{\omega}_1^{(n)} &= \tilde{\mathbf{0}}, \\ \tilde{\omega}_2^{(n)} &= -\tilde{\omega}_0^{(n-1)}, \\ \tilde{\omega}_3^{(n)} &= -\tilde{\omega}_0^{(n-1)} + \tilde{\omega}_2^{(n-1)} - \frac{1}{2} \mathbf{A}_{2,n-1} + \frac{1}{12} [\tilde{\omega}_0^{(n-1)} + \tilde{\omega}_2^{(n-1)}, \mathbf{A}_{2,n-1}], \\ \tilde{\omega}_4^{(n)} &= -\tilde{\omega}_0^{(n-1)} + \tilde{\omega}_3^{(n-1)} - \frac{1}{2} \mathbf{A}_{3,n-1} + \frac{1}{12} [\tilde{\omega}_0^{(n-1)} + \tilde{\omega}_3^{(n-1)}, \mathbf{A}_{3,n-1}]\end{aligned}$$

with $\mathbf{A}_{i,n} := [\tilde{\omega}_0^{(n)}, \tilde{\omega}_i^{(n)}]$, ($i = 2, 3$) and

$$\frac{1}{h} \sum_{i=0}^4 \alpha_i \tilde{\omega}_i^{(n)} = \tilde{\mathbf{v}}_{n+1} + \frac{1}{2} [\tilde{\omega}_0^{(n)}, \tilde{\mathbf{v}}_{n+1}] + \frac{1}{12} [\tilde{\omega}_0^{(n)}, [\tilde{\omega}_0^{(n)}, \tilde{\mathbf{v}}_{n+1}]],$$

i.e.

$$\begin{aligned}\frac{1}{h} \left(\frac{25}{12} \tilde{\omega}_0^{(n)} - \frac{23}{12} \tilde{\omega}_0^{(n-1)} - \frac{4}{3} \tilde{\omega}_2^{(n-1)} + \frac{1}{4} \tilde{\omega}_3^{(n-1)} \right) \\ = \tilde{\mathbf{v}}_{n+1} + \frac{1}{2} [\tilde{\omega}_0^{(n)}, \tilde{\mathbf{v}}_{n+1}] + \frac{1}{6} [\tilde{\omega}_0^{(n)}, \tilde{\mathbf{v}}_{n+1}] + \frac{1}{9h} [\tilde{\omega}_0^{(n-1)} + \tilde{\omega}_2^{(n-1)}, \mathbf{A}_{2,n-1}] \\ - \frac{1}{48h} [\tilde{\omega}_0^{(n-1)} + \tilde{\omega}_3^{(n-1)}, \mathbf{A}_{3,n-1}] - \frac{1}{2h} \mathbf{A}_{2,n-1} + \frac{1}{8h} \mathbf{A}_{3,n-1}.\end{aligned} \quad (27)$$

BLieDF with $k = 4$ may be implemented with the evaluation of one Lie bracket per time step, see (26). The numerical effort of the Munthe-Kaas BDF Lie group method (27) is substantially larger since at least six Lie brackets have to be evaluated and additional updates may be necessary in every Newton step since $\mathbf{v}_{n+1}$ and $\tilde{\omega}_0^{(n)}$ are involved (as for $k = 3$).

6. Convergence analysis

The convergence of multistep methods for differential equations on manifolds is discussed in [9] and proves for $k \leq 6$ the convergence of Munthe-Kaas BDF Lie group integrators with order $p = k$ (in the unconstrained case). In the present section the convergence of fixed step size BLieDF for unconstrained systems is studied in Section 6.1 and for the constrained case in Section 6.2. The convergence of Munthe-Kaas BDF Lie group integrators for constrained systems may be studied in a similar way and will be discussed in full detail in future work. The convergence analysis for variable step size implementations is substantially more complex and requires additional considerations, for example stability conditions [18].

6.1. Convergence analysis for BLieDF: Unconstrained systems

To prove the convergence of a multistep method in linear spaces, the method has to be consistent and zero-stable, see [10]. For consistency, the local truncation error is defined by inserting the analytical solution into the numerical

method. For Eq. (16c) the local truncation errors $\mathbf{I}_n^{\mathbf{v}}$ are given by

$$\mathbf{M}(q(t_{n+1})) \frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{v}(t_{n+1-i}) = -\mathbf{g}(t_{n+1}, q(t_{n+1}), \mathbf{v}(t_{n+1})) + \mathbf{I}_n^{\mathbf{v}}, \quad (28)$$

and correspond to the ones in linear spaces. Therefore, $\mathbf{I}_n^{\mathbf{v}} = \mathcal{O}(h^k)$, see [10]. Inserting the analytical solution $\Omega_{n+1-i}(h) \approx \omega_0^{(n+1-i)}$ into (16a), Eq. (9) for $m = n$ and $t = t_{n+1}$ is obtained and therefore there is no local truncation error in this equation. Finally the local truncation errors $\mathbf{I}_n^{\omega}$ in (16b) are defined by

$$\frac{1}{h} \sum_{i=1}^k \gamma_i \Omega_{n+1-i}(h) = \mathbf{v}(t_{n+1}) + \mathbf{L}_k(\mathbf{v}(t_n), \mathbf{v}(t_{n-1}), \dots, \mathbf{v}(t_{n-k+1}); h) + \mathbf{I}_n^{\omega}, \quad (29)$$

since $\omega_0^{(n+1-i)} \approx \Omega_{n+1-i}(h)$. Lemma 1 implies $\mathbf{I}_n^{\omega} = \mathcal{O}(h^k)$ for $k \leq 4$ if the correction terms $\mathbf{L}_k$ are defined according to (17).

The global error in the configuration variables $q \in G$ may be studied conveniently defining $\tilde{\mathbf{e}}_n^q$ as an element of the corresponding Lie algebra [19], see also [20]. The global errors $\tilde{\mathbf{e}}_n^q \in \mathfrak{g}$, $\mathbf{e}_n^{\omega} \in \mathbb{R}^N$, $\mathbf{e}_n^{\mathbf{v}} \in \mathbb{R}^N$ are defined by

$$q(t_n) = q_n \circ \exp(\tilde{\mathbf{e}}_n^q), \quad \Omega_n(h) = \omega_0^{(n)} + \mathbf{e}_n^{\omega}, \quad \mathbf{v}(t_n) = \mathbf{v}_n + \mathbf{e}_n^{\mathbf{v}}. \quad (30)$$

The convergence analysis is based on the assumption that the numerical solutions q_n and $\omega_0^{(n)}$ remain in a neighbourhood of size $\mathcal{O}(h)$ of the analytical solutions $q(t_n)$ and $\Omega_n(h)$, respectively. Therefore we suppose that there are positive constants h_0 , $\bar{C}$, so that for all $h \in (0, h_0]$ and all r with $t_0 + rh \in [t_0, t_e]$ the global errors are bounded by

$$\|\mathbf{e}_r^q\| + \|\mathbf{e}_r^{\omega}\| \leq \bar{C}h, \quad \|\mathbf{e}_r^{\mathbf{v}}\| \leq \bar{C}. \quad (31)$$

This technical assumption may be verified for BLieDF with $2 \leq k \leq 4$ using techniques that are similar to the ones in the proof of [21, Theorem VII.3.5], see also [20].

For the global errors $\mathbf{e}_n^{\mathbf{v}}$ the difference of (28) and (16b) yields as in linear spaces

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{e}_{n+1-i}^{\mathbf{v}} = \mathcal{O}(1)(\|\mathbf{e}_{n+1}^q\| + \|\mathbf{e}_{n+1}^{\mathbf{v}}\|) + \mathcal{O}(1)\|\mathbf{I}_n^{\mathbf{v}}\| \text{ with } \mathbf{I}_n^{\mathbf{v}} = \mathcal{O}(h^k). \quad (32)$$

The error recursion for the global errors $\mathbf{e}_n^q$ differs from the one in linear configuration spaces.

Theorem 1. The global errors $\mathbf{e}_n^q$ of the BLieDF method (16) satisfy

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{e}_{n+1-i}^q = \mathcal{O}(1) \sum_{i=0}^k (\|\mathbf{e}_{n+1-i}^q\| + \|\mathbf{e}_{n+1-i}^{\mathbf{v}}\|) + \mathbf{I}_n^{\omega} \text{ with } \mathbf{I}_n^{\omega} = \mathcal{O}(h^k). \quad (33)$$

Proof. Inserting (16a) and (9) for $m = n$ and $t = t_{n+1}$ into (30), the relation

$$\begin{aligned} \exp(\tilde{\mathbf{e}}_{n+1}^q) &= q_{n+1}^{-1} \circ q(t_{n+1}) = \exp(-\tilde{\omega}_0^{(n)}) \circ q_n^{-1} \circ q(t_n) \circ \exp(\tilde{\Omega}_n(h)) \\ &= \exp(-\tilde{\omega}_0^{(n)}) \circ \exp(\tilde{\mathbf{e}}_n^q) \circ \exp(\tilde{\Omega}_n(h)) \\ &= \exp(-\tilde{\Omega}_n(h) + \tilde{\mathbf{e}}_n^q) \circ \exp(\tilde{\mathbf{e}}_n^q) \circ \exp(\tilde{\Omega}_n(h)) \end{aligned}$$

holds, see [20]. As a consequence of the Baker–Campbell–Hausdorff formula the estimate

$$\exp(h\tilde{\mathbf{v}}_1) \circ \exp(h\tilde{\mathbf{v}}_2) = \exp(h\tilde{\mathbf{v}}_1 + h\tilde{\mathbf{v}}_2 + \frac{1}{2}[h\tilde{\mathbf{v}}_1, h\tilde{\mathbf{v}}_2] + \mathcal{O}(h)\|[h\tilde{\mathbf{v}}_1, h\tilde{\mathbf{v}}_2]\|) \quad (34)$$

for $h \rightarrow 0$ with $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^N$ can be verified [11]. By applying twice Eq. (34), the composition of exponential maps can be studied:

$$\begin{aligned} \exp(\tilde{\mathbf{e}}_{n+1}^q) &= \exp(-\tilde{\Omega}_n(h) + \tilde{\mathbf{e}}_n^q + \tilde{\mathbf{e}}_n^q - \frac{1}{2}[\tilde{\Omega}_n(h), \tilde{\mathbf{e}}_n^q] + \mathcal{O}(h^2)\|\mathbf{e}_n^q\| + \mathcal{O}(h)\|\mathbf{e}_n^{\omega}\|) \circ \exp(\tilde{\Omega}_n(h)) \\ &= \exp(\tilde{\mathbf{e}}_n^q + \tilde{\mathbf{e}}_n^q + [\tilde{\mathbf{e}}_n^q, \tilde{\Omega}_n(h)] + \mathcal{O}(h^2)\|\mathbf{e}_n^q\| + \mathcal{O}(h)\|\mathbf{e}_n^{\omega}\|), \end{aligned}$$

since higher order terms may be estimated by assumption (31), see [20]. If the argument of the exponential map is considered, equation

$$\tilde{\mathbf{e}}_{n+1}^q = \tilde{\mathbf{e}}_n^q + \tilde{\mathbf{e}}_n^q + [\tilde{\mathbf{e}}_n^q, \tilde{\Omega}_n(h)] + \mathcal{O}(h^2)\|\mathbf{e}_n^q\| + \mathcal{O}(h)\|\mathbf{e}_n^{\omega}\| \quad (35)$$

follows with $\tilde{\Omega}_n(h) = h\tilde{\mathbf{v}}(t_n) + \mathcal{O}(h^2)$ and

$$(\mathbf{I} + \mathcal{O}(h))\mathbf{e}_n^{\omega} = \mathbf{e}_{n+1}^{\omega} - \mathbf{e}_n^q + \mathcal{O}(h)\|\mathbf{e}_n^q\|, \quad (36)$$

i.e., $\mathbf{e}_n^\omega = \mathcal{O}(1)(\|\mathbf{e}_{n+1}^q\| + \|\mathbf{e}_n^q\|)$. By subtracting (16b) from (29), equation

$$\frac{1}{h} \sum_{i=1}^k \gamma_i \mathbf{e}_{n+1-i}^\omega = \mathbf{e}_{n+1}^v + \mathcal{O}(h) \sum_{i=1}^k \|\mathbf{e}_{n+1-i}^v\| + \mathbf{I}_n^\omega \quad (37)$$

is obtained, because of

$$\mathbf{L}_k(\mathbf{v}(t_n), \dots, \mathbf{v}(t_{n-k+1}); h) - \mathbf{L}_k(\mathbf{v}_n, \dots, \mathbf{v}_{n-k+1}; h) = \mathcal{O}(h) \sum_{i=1}^k \|\mathbf{e}_{n+1-i}^v\|.$$

All in all the assertion follows with (35), (36) and (37), since

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{e}_{n+1-i}^q = \frac{1}{h} \sum_{i=1}^k \gamma_i (\mathbf{e}_{n+2-i}^q - \mathbf{e}_{n+1-i}^q) = \frac{1}{h} \sum_{i=1}^k \gamma_i \mathbf{e}_{n+1-i}^\omega + \mathcal{O}(1) \sum_{i=0}^k \|\mathbf{e}_{n+1-i}^q\|. \quad \blacksquare$$

Eqs. (32) and (33) may be summarized to the k -step error recursion

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \begin{bmatrix} \mathbf{e}_{n+1-i}^q \\ \mathbf{e}_{n+1-i}^v \end{bmatrix} = \mathcal{O}(1) \sum_{i=0}^k (\|\mathbf{e}_{n+1-i}^q\| + \|\mathbf{e}_{n+1-i}^v\|) + \mathcal{O}(h^k)$$

for $2 \leq k \leq 4$. Convergence with order $p = k$ follows from the zero-stability of BDF, see [10, Theorem III.4.5].

6.2. Convergence analysis for BLieDF: Constrained systems

The proof of convergence for the BLieDF method for constrained equations of motion (20) is much more complex. The holonomic constraint (18c) and the global errors $\mathbf{e}_n^\lambda$ in the Lagrange multipliers λ must be considered ($\lambda(t_n) = \lambda_n + \mathbf{e}_n^\lambda$). The special structure of the constrained system results in additional higher order terms in all error estimates. Furthermore the first k time steps must be examined separately, which was discussed in [22] in full detail and is omitted in the present paper.

For linear spaces Lötstedt and Petzold [23] showed the convergence of BDF for DAEs of Hessenberg form. For the Lie group integrator, the convergence analysis starts with local error estimates (Theorem 2) and uses difference approximations of hidden constraints to account for constraint equations and Lagrange multipliers (Theorem 4). In the end, a coupled error recursion is obtained by which the convergence order $p = k$ can be proven.

Theorem 2. The local truncation errors of the k -step BLieDF method (20) fulfil for $k \leq 4$

$$\mathbf{I}_n^\omega = \mathcal{O}(h^k), \quad \mathbf{I}_n^v = \mathcal{O}(h^k) \quad \text{and} \quad \left\| \frac{\mathbf{I}_{n+1}^\omega - \mathbf{I}_n^\omega}{h} \right\| = \mathcal{O}(h^k). \quad (38)$$

Proof.

- As for unconstrained systems the local truncation errors $\mathbf{I}_n^\omega$ are defined by (29) and with Lemma 1 the bound $\|\mathbf{I}_n^\omega\| = \mathcal{O}(h^k)$ can be shown. The local truncation errors $\mathbf{I}_n^v$ can be handled as in the unconstrained case and fulfil (38).
- Let $\mathbf{I}_n^\omega$ be the local truncation error for the k -step BLieDF method (20) for the time step $t_n \rightarrow t_{n+1}$. The proof of Lemma 1 and (29) shows, that this error is given by $\mathbf{I}_n^\omega = -\frac{1}{k+1} h^k \mathbf{v}^{(k)}(t_n) + h^k \mathbf{c}(\mathbf{v}(t_n), \dots, \mathbf{v}^{(k-1)}(t_n)) + \mathcal{O}(h^{k+1})$ (see [10, Table III.2.1]) with a function $\mathbf{c}$, that consists of Lie brackets. Therefore the assertion follows, because of

$$\begin{aligned} \frac{\mathbf{I}_{n+1}^\omega - \mathbf{I}_n^\omega}{h} &= \frac{-\frac{1}{k+1} h^k \mathbf{v}^{(k)}(t_{n+1}) + h^k \mathbf{c}(\mathbf{v}(t_{n+1}), \dots, \mathbf{v}^{(k-1)}(t_{n+1}))}{h} \\ &\quad - \frac{-\frac{1}{k+1} h^k \mathbf{v}^{(k)}(t_n) + h^k \mathbf{c}(\mathbf{v}(t_n), \dots, \mathbf{v}^{(k-1)}(t_n))}{h} + \mathcal{O}(h^k) = \mathcal{O}(h^k). \quad \blacksquare \end{aligned}$$

A global error recursion for the variables q and $\mathbf{v}$ is needed to prove the convergence of BLieDF (20). Equation

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{e}_{n+1-i}^v = -[\mathbf{M}^{-1} \mathbf{B}^T](q(t_{n+1})) \mathbf{e}_{n+1}^\lambda + \mathcal{O}(1)(\|\mathbf{e}_{n+1}^q\| + \|\mathbf{e}_{n+1}^v\|) + \mathbf{I}_n^v \quad (39)$$

is immediately obtained from (20c), because this equation is defined in the linear space $\mathbb{R}^N$. The error recursion in q follows similar to the proof of Theorem 1, but here some higher order terms must be considered.

Theorem 3. The global errors $\mathbf{e}_n^q$ of the BLieDF method (20) satisfy for $2 \leq k \leq 4$

$$\begin{aligned} \frac{1}{h} \sum_{i=0}^k \alpha_i \tilde{\mathbf{e}}_{n+1-i}^q &= \tilde{\mathbf{e}}_{n+1}^q + \sum_{i=1}^k \gamma_i [\tilde{\mathbf{e}}_{n+1-i}^q, \tilde{\mathbf{v}}(t_n)] + \tilde{\mathbf{I}}_n^\omega + \mathcal{O}(h) \sum_{i=1}^k \|\Delta_h \mathbf{e}_{n+1-i}^q\| \\ &+ \mathcal{O}(h) \sum_{i=0}^k (\|\mathbf{e}_{n+1-i}^q\| + \|\mathbf{e}_{n+1-i}^v\|) \end{aligned} \quad (40)$$

where $\Delta_h \mathbf{e}_n^q := (\mathbf{e}_{n+1}^q - \mathbf{e}_n^q)/h$.

Proof. The proof starts similar to the proof of Theorem 1 since Eq. (16a) corresponds to (20) and (16b) to (20b). To consider the higher order terms, the last step needs to be adapted and with (35), (36) and (37) equation

$$\begin{aligned} \frac{1}{h} \sum_{i=0}^k \alpha_i \tilde{\mathbf{e}}_{n+1-i}^q &= \frac{1}{h} \sum_{i=1}^k \gamma_i (\tilde{\mathbf{e}}_{n+2-i}^q - \tilde{\mathbf{e}}_{n+1-i}^q) \\ &= \tilde{\mathbf{e}}_{n+1}^q + \frac{1}{h} \sum_{i=1}^k \gamma_i [\tilde{\mathbf{e}}_{n+1-i}^q, \tilde{\Omega}_{n+1-i}(h)] + \tilde{\mathbf{I}}_n^\omega + \mathcal{O}(h) \sum_{i=1}^k (\|\mathbf{e}_{n+1-i}^q\| + \|\mathbf{e}_{n+1-i}^v\|) \\ &+ \mathcal{O}(1) \sum_{i=1}^k \|\mathbf{e}_{n+1-i}^\omega\|. \end{aligned}$$

is obtained.

The assertion follows with $\tilde{\Omega}_{n+1-i}(h) = h\tilde{\mathbf{v}}(t_n) + \mathcal{O}(h^2)$, see (14) and

$$\|\mathbf{e}_{n+1-i}^\omega\| = \mathcal{O}(h) (\|\Delta_h \mathbf{e}_{n+1-i}^q\| + \|\mathbf{e}_{n+1-i}^q\|) \text{ for } i = 1, \dots, k,$$

see (36). ■

All in all the convergence of order $p = k$ of the BLieDF integrator (20) for constrained systems (18c) can be proven with the following theorem.

Theorem 4. If the global errors $\mathbf{e}_r^q$, $\mathbf{e}_r^\omega$, $\mathbf{e}_r^v$ and $\mathbf{e}_r^\lambda$ satisfy the technical assumption

$$\|\mathbf{e}_r^q\| + \|\mathbf{e}_r^\omega\| \leq \bar{C}h, \quad \|\mathbf{e}_r^v\| + \|\mathbf{e}_r^\lambda\| \leq \bar{C}. \quad (41)$$

for a constant $\bar{C} > 0$, all r with $t_0 + rh \in [t_0, t_e]$ and all $h \in (0, h_0]$ and the starting values $q_0, \dots, q_{2k-1}$, $\mathbf{v}_0, \dots, \mathbf{v}_{2k-1}$ and $\lambda_0, \dots, \lambda_{2k-1}$ satisfy

$$\sum_{i=0}^{2k-1} \|\mathbf{e}_i^q\| = \mathcal{O}(h^{k+1}), \quad \sum_{i=0}^{2k-1} \|\mathbf{e}_i^v\| + \|\mathbf{e}_i^\lambda\| = \mathcal{O}(h^k), \quad \max_{0 \leq i \leq 2k-1} \|\Phi(q_i)\| = \mathcal{O}(h^{k+2}), \quad (42)$$

then there are positive constants C , L and h_0 independent of n and h , such that for all $h \in (0, h_0]$ and all $n \geq 0$ with $t_0 + nh \leq t_e - h$ the following global error bounds hold

$$\|\mathbf{e}_n^q\| + \|\mathbf{e}_n^v\| + \|\mathbf{e}_n^\lambda\| \leq Ce^{L(t_n - t_0)} h^k$$

and the BLieDF integrator (20) has the order of convergence $p = k$ for $2 \leq k \leq 4$.

Proof. The complete proof is given in report [24], see also the shorter version in the Appendix. The proof of this theorem includes the following steps:

- An error recursion for $\Delta_h \mathbf{e}_n^q$, which follows directly from (40) is introduced.
- An error estimate for $\|\mathbf{e}_{n+1}^\lambda\|$ must be verified. Therefore the examination of the difference of the holonomic constraints (20d) at different time points is important to get an approximation of

$$\frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{e}_{n+1-i}^v = \frac{1}{h} \sum_{i=1}^k \gamma_i (\mathbf{e}_{n+2-i}^v - \mathbf{e}_{n+1-i}^v)$$

which can be inserted into (39). The details are very technical and include e.g. the result of part (a), the derivatives of the holonomic constraints (20d), the local error estimates (38), as well as the technical assumptions (41). Furthermore this error estimate of $\|\mathbf{e}_{n+1}^\lambda\|$ can only be proven for $n \geq 2k - 1$ in this way. For the first k time steps ($n < 2k$) another strategy must be applied which was described in [22].

- A coupled error recursion is obtained by combining parts (a), (b) and Eqs. (39) and (40) that allows to prove the order of convergence $p = k$, see also [13]. ■

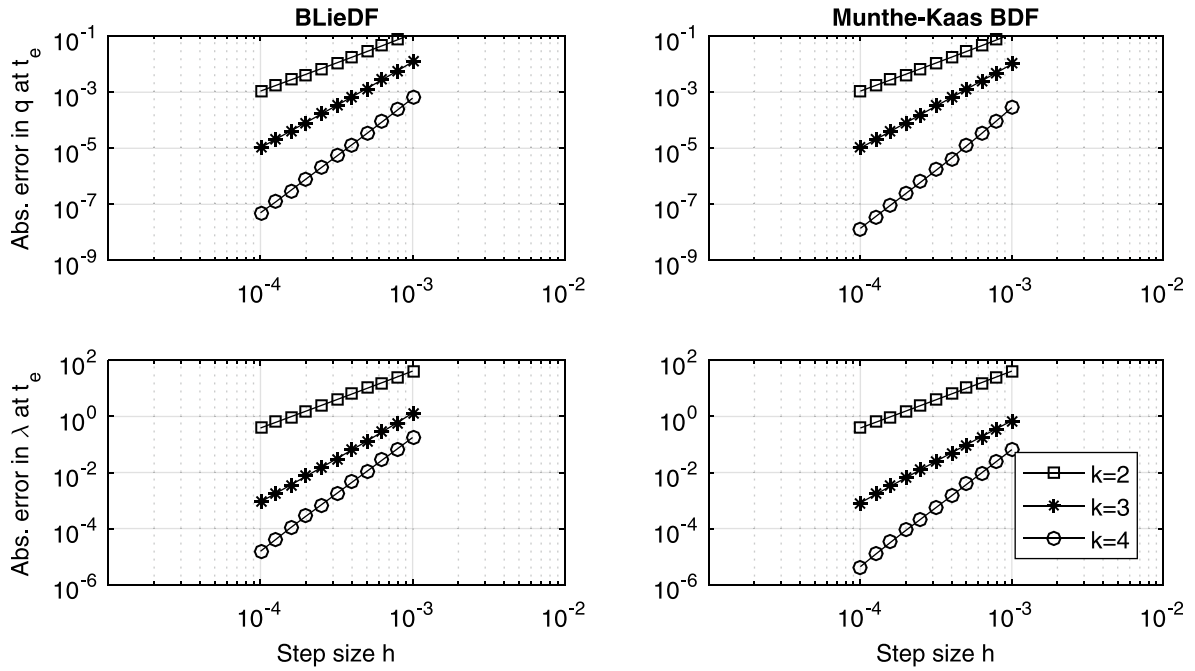


Fig. 3. Global errors for the Heavy top benchmark for BDF integrators in Lie group formulation (Upper left plot: Integrator (20) – absolute error in q at t_e . Upper right plot: Integrator (23) – absolute error in q at t_e . Lower left plot: Integrator (20) – absolute error in λ at t_e . Lower right plot: Integrator (23) – absolute error in λ at t_e).

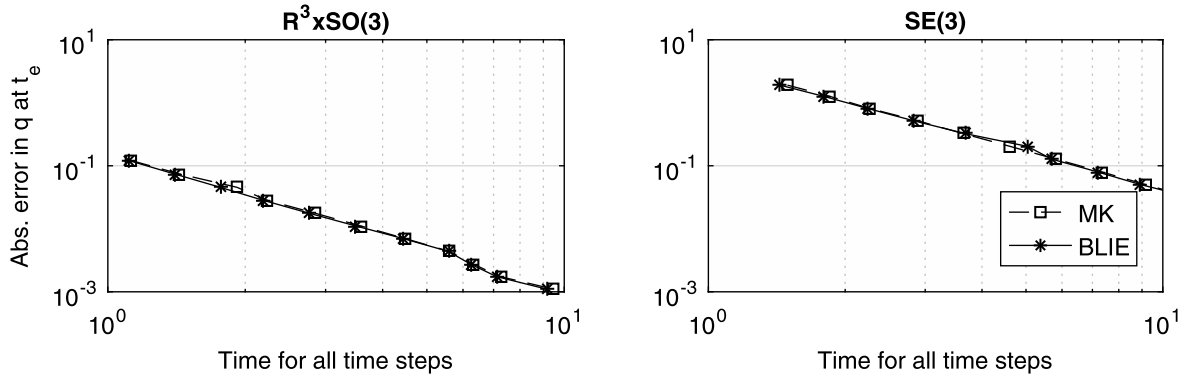


Fig. 4. Efficiency of the integrators (20) and (22)/(23) for $k=2$ in Heavy to benchmark (Left plot: $\mathbb{R}^3 \times SO(3)$. Right plot: $SE(3)$).

7. Numerical tests

The theoretical results should be verified in the numerical tests. Both integrators (20), (22a)–(22c), (23) are compared in terms of order of convergence and efficiency. For the computing time the influence of implementation and Lie group formulation should be examined. The numerical tests are done for the Heavy top benchmark, see Fig. 5, which is a rotating spinning top with its tip fixed at the origin. The Lie group formulations $\mathbb{R}^3 \times SO(3)$ and $SE(3) = \mathbb{R}^3 \ltimes SO(3)$ with the special orthogonal group $SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} : \mathbf{R}^T \mathbf{R} = \mathbf{I}_3, \det \mathbf{R} = 1\}$ are examined [25]. At first the order of convergence was tested for the Heavy top benchmark in the Lie group formulation $\mathbb{R}^3 \times SO(3)$ in Fig. 3. The absolute error at t_e of position coordinate q (upper plots) and Lagrange multipliers λ (lower plots) over the step size h is plotted in double logarithmic scale for the BLieDF integrator (left plots) and the Munthe-Kaas BDF Lie group integrator (right plots). The numerical test illustrates an order of convergence of size $p = k$ for $2 \leq k \leq 4$ for both integrators for the solution components q and λ . This corresponds to the theoretical examinations in Section 6.

For the efficiency both integrators are tested for $k=2$ at first. Section 5 shows that there is no difference between the two integrators in this case. If implementation aspects have no influence on the computing time, then both integrators should have the same behaviour. In Fig. 4 the absolute error at position coordinate q is plotted over the computing time

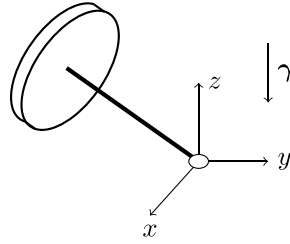
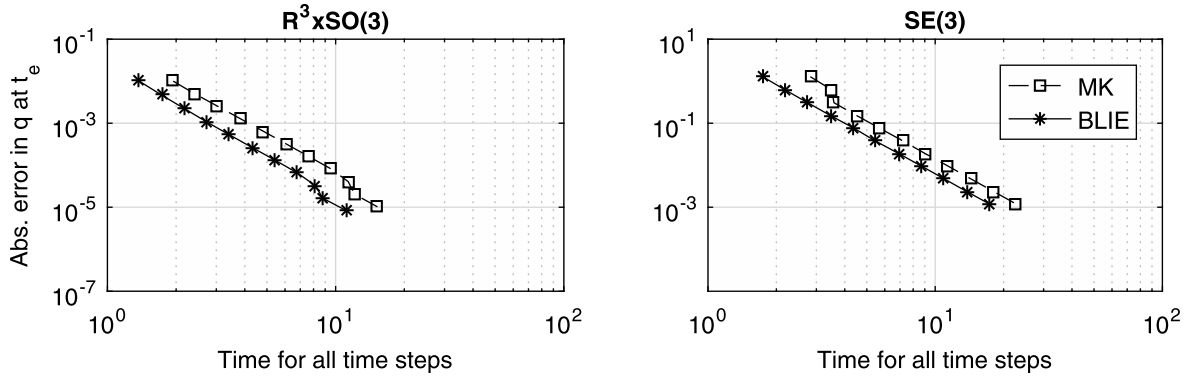
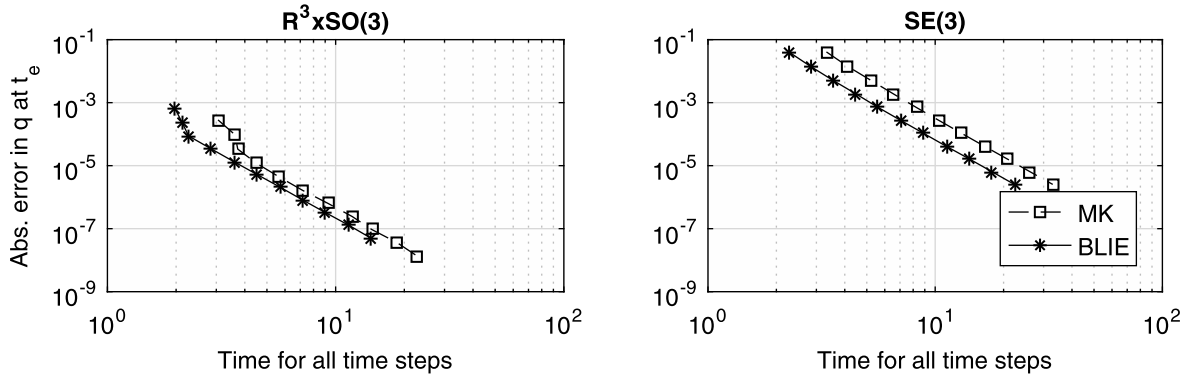


Fig. 5. Heavy top.

Fig. 6. Efficiency of the integrators (24) and (25) for $k = 3$ in Heavy to benchmark (Left plot: $\mathbb{R}^3 \times SO(3)$. Right plot: $SE(3)$).Fig. 7. Efficiency of the integrators (26) and (27) for $k = 4$ in Heavy to benchmark (Left plot: $\mathbb{R}^3 \times SO(3)$. Right plot: $SE(3)$).

that is needed for all time steps without initialization for the Heavy top benchmark in the Lie group formulations $\mathbb{R}^3 \times SO(3)$ (left plots) and $SE(3)$ (right plots). This test confirms, that there is no difference between the two integrators for $k = 2$.

The three-step integrators (24) and (25) are tested in Fig. 6. The absolute error at position coordinate q is plotted over the computing time that is needed for all time steps without the time for initialization for the Heavy top benchmark in the Lie group formulations $\mathbb{R}^3 \times SO(3)$ (left plots) and $SE(3)$ (right plots). To reach the same global error the BLieDF method needs less computing time than the Munthe-Kaas BDF Lie group integrator regardless of which Lie group formulation is used.

Similar test results can be observed if the four-step integrators (26) and (27) are used, see Fig. 7.

The numerical tests show that the BLieDF integrator was superior to the Munthe-Kaas BDF Lie group integrator independent of the Lie group formulation and implementation aspects.

8. Conclusions

The BLieDF integrator was introduced for unconstrained and constrained mechanical systems in Lie group formulations. The equations of motion form ODEs or index-3 DAEs on Lie groups. In both cases, the convergence analysis proves an

order of convergence of $p = k$ for $2 \leq k \leq 4$ for BLieDF. The novel integrator was compared with an earlier approach of Falinsen, Marthinsen and Munthe-Kaas and it was shown, that for $3 \leq k \leq 4$ the BLieDF method is more efficient, since it calculates less Lie brackets. For $k = 2$ both integrators are equivalent. The results were confirmed by numerical test results for the Heavy top benchmark in the Lie group formulations $\mathbb{R}^3 \times SO(3)$ and $SE(3)$.

Appendix. Proof of Theorem 4

Lemma A.1. The global errors $\Delta_h \mathbf{e}_n^q = (\mathbf{e}_{n+1}^q - \mathbf{e}_n^q)/h$ of the BLieDF integrator (20) for $k \geq 2$ satisfy

$$\begin{aligned} \sum_{i=1}^k \gamma_i \Delta_h \mathbf{e}_{n+1-i}^q &= \mathbf{e}_{n+1}^q - \sum_{i=1}^k \gamma_i \widehat{\mathbf{v}}(t_n) \mathbf{e}_{n+1-i}^q + \mathcal{O}(h) \sum_{i=1}^k \|\Delta_h \mathbf{e}_{n+1-i}^q\| \\ &+ \mathcal{O}(h) \sum_{i=0}^k (\|\mathbf{e}_{n+1-i}^q\| + \|\mathbf{e}_{n+1-i}^v\|) + \mathbf{I}_n^\omega \end{aligned} \quad (\text{A.1})$$

with the linear operator $(\widehat{\bullet}) : \mathbb{R}^N \rightarrow \mathbb{R}^{N \times N}$, $\mathbf{v} \mapsto \widehat{\mathbf{v}}$ that is defined by $\widehat{\mathbf{v}}\mathbf{w} = [\widehat{\mathbf{v}}, \mathbf{w}]$ for all $\mathbf{w} \in \mathbb{R}^N$.

Proof. Writing (40) in terms of γ_i for $i = 1, \dots, k$, the assertion follows from Theorem 3, since

$$\sum_{i=1}^k \gamma_i \Delta_h \mathbf{e}_{n+1-i}^q = \sum_{i=1}^k \gamma_i \frac{\mathbf{e}_{n+2-i}^q - \mathbf{e}_{n+1-i}^q}{h} = \frac{1}{h} \sum_{i=0}^k \alpha_i \mathbf{e}_{n+1-i}^q. \quad \blacksquare$$

Lemma A.2. The global errors $\mathbf{e}_{n+1}^{S\lambda}$ of the BLieDF integrator (20) for $2 \leq k \leq 4$ satisfy for $n \geq 2k - 1$

$$\|\mathbf{e}_{n+1}^{S\lambda}\| = \mathcal{O}\left(\frac{1}{h^2}\right) \max_r \|\Phi(q_r)\| + \mathcal{O}(1) \left(\sum_{i=0}^{2k} (\|\mathbf{e}_{n+1-i}^q\| + \|\mathbf{e}_{n+1-i}^v\|) + \sum_{i=1}^{2k} \|\Delta_h \mathbf{e}_{n+1-i}^q\| \right) + \mathcal{O}(h^k) \quad (\text{A.2})$$

with the notation $\mathbf{e}^{(\bullet)} := \mathbf{C}(q(t_n))\mathbf{e}_n^{(\bullet)}$ and $\mathbf{S} := \mathbf{B}\mathbf{M}^{-1}\mathbf{B}^T$.

Proof. By multiplying (39) from the left with $\mathbf{B}(q(t_{n+1}))$ and rewriting the sum in terms of γ_i , equation

$$\sum_{i=1}^k \gamma_i \frac{\mathbf{e}_{n+2-i}^{Bv} - \mathbf{e}_{n+1-i}^{Bv}}{h} = -\mathbf{e}_{n+1}^{S\lambda} + \mathcal{O}(1) \left(\|\mathbf{e}_{n+1}^q\| + \sum_{i=0}^k \|\mathbf{e}_{n+1-i}^v\| \right) + \mathcal{O}(h^k) \quad (\text{A.3})$$

follows with Theorem 2. Note, that $\mathbf{B}(q(t_{n+1})) = \mathbf{B}(q(t_{n+1-i})) + \mathcal{O}(h)$ for $i = 0, \dots, k$.

Furthermore with (18c) and defining $q_{n,\vartheta} := q(t_n) \circ \exp(-\vartheta \mathbf{e}_n^q)$, the relation

$$-\sum_{j=1}^k \gamma_j \Phi(q_{n+1-j}) = -\sum_{j=1}^k \gamma_j (\Phi(q_{n+1-j,1}) - \Phi(q_{n+1-j,0})) = \sum_{j=1}^k \gamma_j \int_0^1 \mathbf{B}(q_{n+1-j,\vartheta}) \mathbf{e}_{n+1-j}^q d\vartheta$$

holds, see (19) and [20]. By scaling the difference of this sum for $t = t_{n+2-i}$ and $t = t_{n+1-i}$ by $\frac{1}{h}$, equation

$$-\sum_{j=1}^k \gamma_j \frac{\Phi(q_{n+2-j}) - \Phi(q_{n+1-j})}{h} = \sum_{j=1}^k \gamma_j \int_0^1 \frac{\mathbf{B}(q_{n+2-j,\vartheta}) \mathbf{e}_{n+2-j}^q - \mathbf{B}(q_{n+1-j,\vartheta}) \mathbf{e}_{n+1-j}^q}{h} d\vartheta$$

is valid. Proceeding as in the proof of [20, Lemma 4], see [20, Eq. (27)], we get the estimate

$$\begin{aligned} -\sum_{j=1}^k \gamma_j \frac{\Phi(q_{n+2-j}) - \Phi(q_{n+1-j})}{h} &= \mathbf{B}(q(t_n)) \sum_{j=1}^k \gamma_j \Delta_h \mathbf{e}_{n+1-j}^q + \mathbf{R}(q(t_n)) \left(\sum_{j=1}^k \gamma_j \mathbf{e}_{n+1-j}^q, \mathbf{v}(t_n) \right) \\ &+ \mathcal{O}(h) \sum_{j=1}^k (\|\mathbf{e}_{n+1-j}^q\| + \|\Delta_h \mathbf{e}_{n+1-j}^q\|) \end{aligned} \quad (\text{A.4})$$

with a bilinear form $\mathbf{R} : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}^l$, that is defined by the derivative of $\Theta(q, \mathbf{w}) := \mathbf{B}(q)\mathbf{w}$ with respect to q in the sense that

$$D_q \Theta(q, \mathbf{w}) \cdot (DL_q(e) \cdot \tilde{\mathbf{z}}) = \mathbf{R}(q)(\mathbf{w}, \mathbf{z}) \text{ for all } \mathbf{w}, \mathbf{z} \in \mathbb{R}^N.$$

Taking (A.1) into account, the first term at the right hand side of (A.4) is expressed as

$$\begin{aligned} \mathbf{B}(q(t_n)) \sum_{j=1}^k \gamma_j \Delta_h \mathbf{e}_{n+1-j}^q &= \mathbf{e}_{n+1}^{\mathbf{Bv}} - \mathbf{B}(q(t_n)) \widehat{\mathbf{v}}(t_n) \sum_{j=1}^k \gamma_j \mathbf{e}_{n+1-j}^q + \mathbf{B}(q(t_n)) \mathbf{l}_n^\omega \\ &+ \mathcal{O}(h) \left(\sum_{j=0}^k (\|\mathbf{e}_{n+1-j}^q\| + \|\mathbf{e}_{n+1-j}^{\mathbf{v}}\|) + \sum_{j=1}^k \|\Delta_h \mathbf{e}_{n+1-j}^q\| \right) \end{aligned}$$

resulting in

$$\begin{aligned} \mathbf{e}_{n+1}^{\mathbf{Bv}} &= - \sum_{j=1}^k \gamma_j \frac{\Phi(q_{n+2-j}) - \Phi(q_{n+1-j})}{h} - \mathbf{B}(q(t_n)) \mathbf{l}_n^\omega + \mathbf{B}(q(t_n)) \widehat{\mathbf{v}}(t_n) \sum_{j=1}^k \gamma_j \mathbf{e}_{n+1-j}^q \\ &- \mathbf{R}(q(t_n)) \left(\sum_{j=1}^k \gamma_j \mathbf{e}_{n+1-j}^q, \mathbf{v}(t_n) \right) + \mathcal{O}(h) \left(\sum_{j=0}^k (\|\mathbf{e}_{n+1-j}^q\| + \|\mathbf{e}_{n+1-j}^{\mathbf{v}}\|) + \sum_{j=1}^k \|\Delta_h \mathbf{e}_{n+1-j}^q\| \right). \end{aligned} \quad (\text{A.5})$$

The difference of Eq. (A.5) for two time steps scaled by $\frac{1}{h}$ amounts to

$$\begin{aligned} \frac{\mathbf{e}_{n+2-i}^{\mathbf{Bv}} - \mathbf{e}_{n+1-i}^{\mathbf{Bv}}}{h} &= \mathcal{O} \left(\frac{1}{h^2} \right) \max_r \|\Phi(q_r)\| - \mathbf{B}(q(t_n)) \frac{\mathbf{l}_{n+1-i}^\omega - \mathbf{l}_{n-i}^\omega}{h} + \mathcal{O}(h^k) \\ &+ \mathbf{B}(q(t_n)) \widehat{\mathbf{v}}(t_n) \sum_{j=1}^k \gamma_j \Delta_h \mathbf{e}_{n-i+1-j}^q - \mathbf{R}(q(t_n)) \left(\sum_{j=1}^k \gamma_j \Delta_h \mathbf{e}_{n-i+1-j}^q, \mathbf{v}(t_n) \right) \\ &+ \mathcal{O}(1) \left(\sum_{j=0}^k (\|\mathbf{e}_{n+1-j}^q\| + \|\mathbf{e}_{n+1-j}^{\mathbf{v}}\|) + \sum_{j=1}^k \|\Delta_h \mathbf{e}_{n+1-j}^q\| \right). \end{aligned} \quad (\text{A.6})$$

Lemma A.1 shows for all n with $n - i \geq k$

$$\sum_{j=1}^k \gamma_j \Delta_h \mathbf{e}_{n-i+1-j}^q = \mathcal{O}(1) \sum_{j=0}^k (\|\mathbf{e}_{n-i+1-j}^q\| + \|\mathbf{e}_{n-i+1-j}^{\mathbf{v}}\|) + \mathcal{O}(h^k), \quad (\text{A.7})$$

such that the assertion of the lemma follows from the local error estimate (38) if (A.6) and (A.7) are substituted into (A.3). ■

Proof of Theorem 4. If $0 \leq n < 2k - 1$, then the assertion follows from assumptions (42). For $n \geq 2k - 1$ a one step error recursion can be obtained by combining the estimates (39), (40), (A.1) and (A.2) with (38) to

$$\begin{aligned} \|\mathbf{E}_{n+1}^{\mathbf{y}} - \mathbf{T}_{\mathbf{y}} \mathbf{E}_n^{\mathbf{y}}\| &\leq \mathcal{O}(h) (\|\mathbf{E}_{n+1}^{\mathbf{y}}\| + \|\mathbf{E}_n^{\mathbf{y}}\| + \|\mathbf{E}_{n+1}^{\mathbf{z}}\| + \|\mathbf{E}_n^{\mathbf{z}}\|) + \mathcal{O}(h^{k+1}), \\ \|\mathbf{E}_{n+1}^{\mathbf{z}} - \mathbf{T}_{\mathbf{z}} \mathbf{E}_n^{\mathbf{z}}\| &\leq \mathcal{O}(1) (\|\mathbf{E}_{n+1}^{\mathbf{y}}\| + h \|\mathbf{E}_{n+1}^{\mathbf{z}}\|) + \mathcal{O}(h^k) \end{aligned}$$

with

$$\mathbf{E}_n^{\mathbf{y}} := \begin{bmatrix} \mathbf{e}_n^q \\ \vdots \\ \mathbf{e}_{n+1-2k}^q \\ \mathbf{e}_n^{\mathbf{v}} \\ \vdots \\ \mathbf{e}_{n+1-2k}^{\mathbf{v}} \end{bmatrix}, \quad \mathbf{E}_n^{\mathbf{z}} := \begin{bmatrix} \Delta_h \mathbf{e}_{n-1}^q \\ \vdots \\ \Delta_h \mathbf{e}_{n+1-2k}^q \end{bmatrix}, \quad \mathbf{T}_{\mathbf{y}} := \begin{bmatrix} \mathbf{T}_\alpha & \mathbf{0}_{2k \times 2k} \\ \mathbf{0}_{2k \times 2k} & \mathbf{T}_\alpha \end{bmatrix} \otimes \mathbf{I}_N, \quad \mathbf{T}_{\mathbf{z}} := \mathbf{T}_{\mathbf{y}} \otimes \mathbf{I}_N,$$

and

$$\begin{aligned} \mathbf{T}_\alpha &= (t_{ij}^\alpha)_{i,j=1,\dots,2k} \text{ with } t_{ij}^\alpha = \begin{cases} -\frac{\alpha_j}{\alpha_0} & \text{if } i = 1, j = 1, \dots, k, \\ 1 & \text{if } i = j + 1, j = 1, \dots, 2k - 1, \\ 0 & \text{otherwise} \end{cases}, \\ \mathbf{T}_{\mathbf{y}} &= (t_{ij}^\gamma)_{i,j=1,\dots,2k-1} \text{ with } t_{ij}^\gamma = \begin{cases} -\frac{\gamma_{j+1}}{\gamma_1} & \text{if } i = 1, j = 1, \dots, k - 1, \\ 1 & \text{if } i = j + 1, j = 1, \dots, 2k - 2, \\ 0 & \text{otherwise} \end{cases}. \end{aligned}$$

Because of (42) and $\Phi(q_r) = \mathbf{0}$ for $r \geq k$, see (20d), the term $\frac{1}{h^2} \max_r \|\Phi(q_r)\|$ in (A.2) is $\mathcal{O}(h^k)$.

Furthermore, a positive constant L independent of h and n and an appropriate constant $M \geq 0$, representing the local error terms $\mathcal{O}(h^k)$, can be chosen, such that for $n \geq 2k - 1$, the coupled error recursion

$$\begin{aligned}\|\mathbf{E}_{n+1}^y - \mathbf{T}_y \mathbf{E}_n^y\| &\leq Lh(\|\mathbf{E}_{n+1}^y\| + \|\mathbf{E}_n^y\| + \|\mathbf{E}_{n+1}^z\| + \|\mathbf{E}_n^z\|) + hM, \\ \|\mathbf{E}_{n+1}^z - \mathbf{T}_z \mathbf{E}_n^z\| &\leq L(\|\mathbf{E}_{n+1}^y\| + h\|\mathbf{E}_{n+1}^z\|) + M\end{aligned}$$

follows. There is a norm $\|\cdot\|_y$ with $\|\mathbf{T}_y\|_y = 1$ (see [10, Lemma III.4.4]). For $k \leq 6$, BDF are zero-stable and the roots of the characteristic polynomial $p_\alpha(\zeta) = \sum_{i=0}^k \alpha_i \zeta^{k-i}$ satisfy $\zeta_1 = 1$ and $|\zeta_i| < 1$, ($i = 2, \dots, k$), see [10]. Because of $\det(\zeta \mathbf{I}_{2k-1} - \mathbf{T}_y) = \zeta^k \sum_{i=1}^k \gamma_i \zeta^{k-i} / \gamma_1$ and

$$(\zeta - 1) \sum_{i=1}^k \gamma_i \zeta^{k-i} = \sum_{i=1}^k \gamma_i (\zeta^{k+1-i} - \zeta^{k-i}) = \sum_{i=0}^k \alpha_i \zeta^{k-i} = p_\alpha(\zeta),$$

all eigenvalues of $\mathbf{T}_y$ are inside the unit circle and there exists a norm $\|\cdot\|_y$ with $\|\mathbf{T}_y\|_y < 1$. With [13, Theorem 4.16] the estimates

$$\begin{aligned}\|\mathbf{E}_n^y\| &\leq \bar{C} e^{\bar{L}(t_n - t_0)} (\|\mathbf{E}_{2k-1}^y\| + h\|\mathbf{E}_{2k-1}^z\| + M), \\ \|\mathbf{E}_{n+1}^z - \mathbf{T}_z^{n+2-2k} \mathbf{E}_{2k-1}^z\| &\leq \bar{C} e^{\bar{L}(t_n - t_0)} (\|\mathbf{E}_{2k-1}^y\| + h\|\mathbf{E}_{2k-1}^z\| + M)\end{aligned}$$

are obtained for step size $h \in (0, h_0]$ and $t_n = t_0 + nh$, ($n \geq 2k - 1$) and positive constants $h_0, \bar{C}, \bar{L}$, that depend on L as well as on the chosen norms. With (42), the estimates $\|\mathbf{E}_{2k-1}^y\| = \mathcal{O}(h^k)$ and $\|\mathbf{E}_{2k-1}^z\| = \mathcal{O}(h^k)$ are obtained. Therefore the assertion follows for $n \geq 2k - 1$ together with Eq. (A.2). ■

References

- [1] Martin Arnold, Bernhard Burgermeister, Claus Führer, Gerhard Hippmann, Georg Rill, Numerical methods in vehicle system dynamics: State of the art and current developments, *Veh. Syst. Dynam.* 49 (2011) 1159–1207.
- [2] Charles Francis Curtiss, Joseph Oakland Hirschfelder, Integration of stiff equations, *Proc. Natl. Acad. Sci. USA* 38 (1952) 235–243.
- [3] Charles William Gear, *Numerical Initial Value Problems in Ordinary Differential Equations*, Prentice-Hall, Englewood Cliffs, NJ, 1971.
- [4] Peter E. Crouch, Robert Grossmann, Numerical integration of ordinary differential equations on manifolds, *J. Nonlinear Sci.* 3 (1992) 1–33.
- [5] Hans Munthe-Kaas, Runge-Kutta methods on Lie groups, *BIT* 38 (1998) 92–111.
- [6] Hans Munthe-Kaas, High order Runge-Kutta methods on manifolds, *Appl. Numer. Math.* 29 (1999) 115–127.
- [7] Elena Celledoni, Arne Marthinsen, Brynjulf Owren, Commutator-free Lie group methods, *Future Gener. Comput. Syst.* 19 (2003) 341–352.
- [8] Olivier Brüls, Alberto Cardona, On the use of Lie group time integrators in multibody dynamics, *J. Comput. Nonlinear Dynam.* 5 (2010) 031002.
- [9] Stig Faltinsen, Arne Marthinsen, Hans Munthe-Kaas, Multistep methods integrating ordinary differential equations on manifolds, *Appl. Numer. Math.* 39 (2001) 349–365.
- [10] Ernst Hairer, Syvert Nørsett, Gerhard Wanner, *Solving Ordinary Differential Equations. I. Nonstiff Problems*, second ed., Springer-Verlag, Berlin Heidelberg New York, 1993.
- [11] Ernst Hairer, Christian Lubich, Gerhard Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*, second ed., Springer-Verlag, Berlin Heidelberg New York, 2006.
- [12] Veeravallis S. Varadarajan, Lie Groups, Lie Algebras, and their Representations, Springer Verlag, New York, Berlin, Heidelberg, Tokyo, 1992.
- [13] Martin Arnold, Alberto Cardona, Olivier Brüls, A Lie algebra approach to Lie group time integration of constrained systems, in: P. Betsch (Ed.), *Structure-Preserving Integrators in Nonlinear Structural Dynamics and Flexible Multibody Dynamics*, Springer International Publishing, Cham, 2016, pp. 91–158.
- [14] Wilhelm Magnus, On the exponential solution of differential equations for a linear operator, *Comm. Pure Appl. Math.* 7 (1954) 649–673.
- [15] Andreas Müller, Approximation of finite rigid body motions from velocity fields, *ZAMM - J. Appl. Math. Mech. / Z. Angew. Math. Mech.* 90 (2010) 514–521.
- [16] Olivier Brüls, Alberto Cardona, Martin Arnold, Lie group generalized- α time integration of constrained flexible multibody systems, *Mech. Mach. Theory* 48 (2012) 121–137.
- [17] Hans Munthe-Kaas, Brynjulf Owren, Computations in a free Lie algebra, *Philos. Trans. R. Soc. Lond.* 357 (1999).
- [18] Victoria Wieloch, Martin Arnold, Stability bounds for step size ratios in variable time step implementations of Newmark integrators, in: *Proceedings of ECCOMAS Thematic Conference on Multibody Dynamics*, Prague, 2017.
- [19] Jörg Wensch, Extrapolation methods in Lie groups, *Numer. Math.* 89 (2001) 591–604.
- [20] Martin Arnold, Olivier Brüls, Alberto Cardona, Error analysis of generalized- α Lie group time integration methods for constrained mechanical systems, *Numer. Math.* 129 (2015) 149–179.
- [21] Ernst Hairer, Gerhard Wanner, *Solving Ordinary Differential Equations. II. Stiff and Differential-Algebraic Problems*, second ed., Springer-Verlag, Berlin Heidelberg New York, 1996.
- [22] Victoria Wieloch, Martin Arnold, BLieDF2nd - a k -step BDF integrator for constrained mechanical systems on Lie groups, in: *Proceedings of the 5th Joint International Conference on Multibody System Dynamics*, Lisbon, 2018.
- [23] Per Lötstedt, Linda Petzold, Numerical solution of nonlinear differential equations with algebraic constraints I: Convergence results for backward differentiation formulas, *Math. Comp.* 46 (174) (1986) 491–516.
- [24] Victoria Wieloch, Martin Arnold, Convergence analysis of BLieDF integrators for constrained mechanical systems. Technical report, 2019 (in preparation).
- [25] Olivier Brüls, Martin Arnold, Alberto Cardona, Two Lie group formulations for dynamic multibody systems with large rotations, in: *Proceedings of the ASME 2011 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference*, 2011.